

Power BI Assignment 1 – Data Transformation, Data Modeling

Transformation Steps :

- ❖ First, I Imported the CSV File Subsequently, I selected “Transform Data” to Load it into Power Query.
- ❖ I then Applied Row limiting and Formatted the data. Following that, I converted the “Customer Name” Field to Proper Case.
- ❖ I Created a New Column by Matching the “City” and “State” Fields, And Generated a Conditional Column Based On The “Profit States” Data.
- ❖ Location New Column

The screenshot shows the Power BI Query Editor with the 'List of Orders' query selected. The formula bar contains the M code: `Table.AddColumn(#"CustomerName to Proper Case", "Location", each Text.Combine({City}, [State]), ", ", type text)`. The 'Applied Steps' pane on the right shows the sequence: Source, Navigation, Promoted Headers, Order Date Data Type Changed, CustomerName to Proper Case, and Merged City and State. The main data view displays columns for Order Date, Customer Name, State, City, and the newly created Location column, which combines City and State separated by a comma.

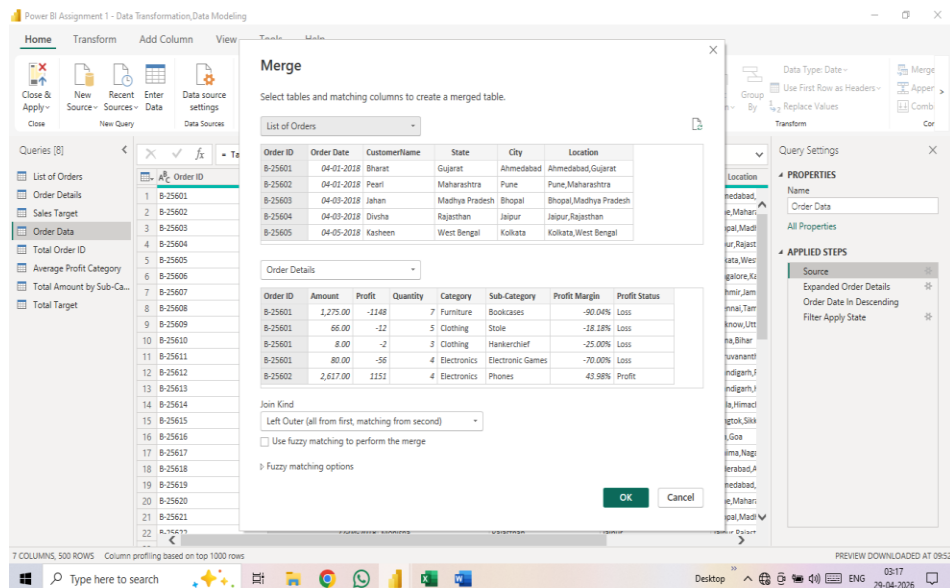
Order Date	Customer Name	State	City	Location
04-01-2018	Bharat	Gujarat	Ahmedabad	Ahmedabad,Gujarat
04-01-2018	Pearl	Maharashtra	Pune	Pune,Maharashtra
04-01-2018	Jahan	Madhya Pradesh	Bhopal	Bhopal,Madhya Pradesh
04-01-2018	Dhivha	Rajasthan	Jaipur	Jaipur,Rajasthan
04-01-2018	Kasheen	West Bengal	Kolkata	Kolkata,West Bengal
04-06-2018	Hazel	Karnataka	Bangalore	Bangalore,Karnataka
04-06-2018	Sonakshi	Jammu and Kashmir	Kashmir	Kashmir,Jammu and Kashmir
04-08-2018	Aarushi	Tamil Nadu	Chennai	Chennai,Tamil Nadu
04-09-2018	Itesh	Uttar Pradesh	Lucknow	Lucknow,Uttar Pradesh
04-09-2018	Yogesh	Bihar	Patna	Patna,Bihar
04-11-2018	Anita	Kerala	Thiruvananthapuram	Thiruvananthapuram,Kerala
04-12-2018	Divachand	Punjab	Chandigarh	Chandigarh,Punjab
04-12-2018	Mukesh	Haryana	Chandigarh	Chandigarh,Haryana
12-04-2018	Vandana	Himachal Pradesh	Simla	Simla,Himachal Pradesh
15-04-2018	Shweta	Sikkim	Gangtok	Gangtok,Sikkim
15-04-2018	Kanak	Goa	Goa	Goa,Goa
17-04-2018	Sagar	Nagaland	Kohima	Kohima,Nagaland
18-04-2018	Manju	Andhra Pradesh	Hyderabad	Hyderabad,Andhra Pradesh
18-04-2018	Ramesh	Gujarat	Ahmedabad	Ahmedabad,Gujarat
20-04-2018	Santa	Maharashtra	Pune	Pune,Maharashtra
20-04-2018	Deepak	Madhya Pradesh	Bhopal	Bhopal,Madhya Pradesh
21-04-2018	Mukesh	Delhi	New Delhi	New Delhi,Delhi

❖ Conditional Column

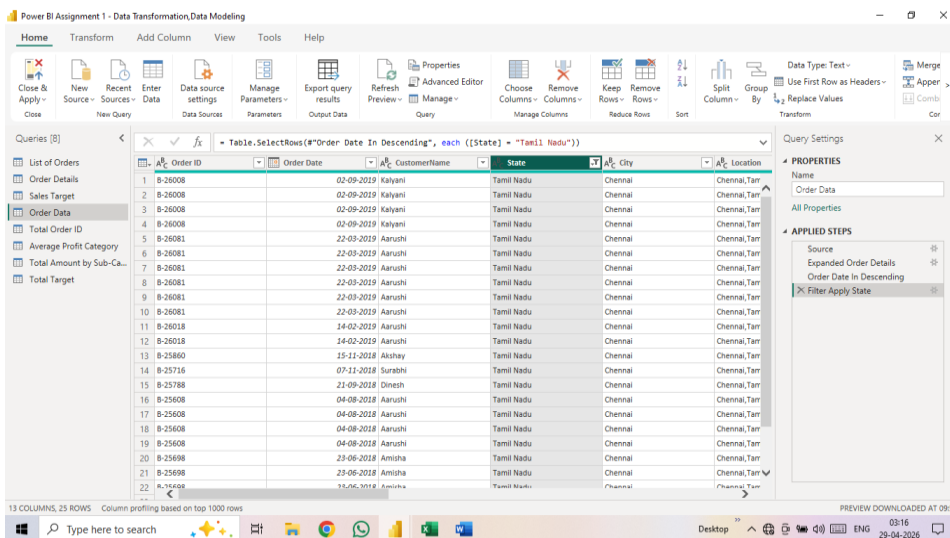
The screenshot shows the Power BI Query Editor with the 'Order Details' query selected. The formula bar contains the M code: `Table.AddColumn(#"Profit Margin Column Data Type Changed", "Profit Status", each if [Profit] < 100 then "Loss"`. The 'Applied Steps' pane on the right shows the sequence: Source, Navigation, Promoted Headers, Changed Data Type, Amount Column Data Type C..., Profit Margin Column Data Ty..., and Profit Status Conditional Colu... The main data view displays columns for Quantity, Category, Sub-Category, Profit Margin, and the newly created Profit Status column, which categorizes orders as 'Loss' based on the profit margin.

Quantity	Category	Sub-Category	Profit Margin	Profit Status
2249	Furniture	Bookcases	-60.04%	Loss
12	Clothing	Stole	-18.18%	Loss
2	Clothing	Hankchief	-25.00%	Loss
56	Electronics	Electronic Games	-70.00%	Loss
111	Electronics	Phones	-66.07%	Loss
272	Electronics	Phones	-64.15%	Loss
111	Electronics	Phones	43.98%	Profit
222	Clothing	Saree	37.79%	Break-Even
5	Clothing	Saree	-4.20%	Loss
40	Clothing	Trousers	-4.43%	Loss
30	Furniture	Chairs	-125.00%	Loss
186	Clothing	Kurti	-86.01%	Loss
5	Clothing	Trousers	47.37%	Loss
16	Clothing	Stole	32.79%	Loss
36	Clothing	Stole	33.64%	Loss
1	Clothing	Hankchief	8.33%	Loss
18	Clothing	T-shirt	26.15%	Loss
17	Clothing	T-shirt	3.18%	Loss
5	Clothing	Saree	0.00%	Loss
0	Clothing	Saree	4.60%	Loss
4	Clothing	Stole	81.96%	Loss

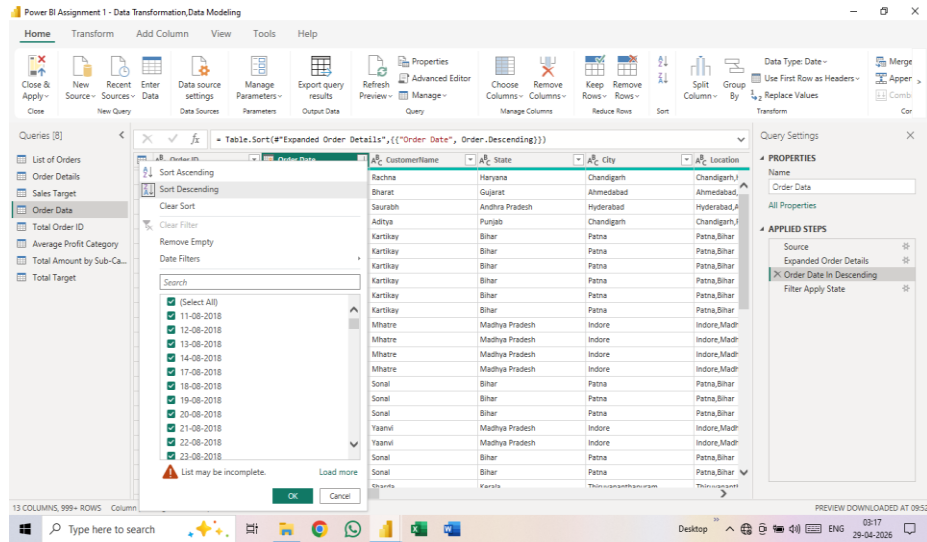
- ❖ Furthermore, I Matched the “List of Orders” and “Order Details” tables to create a Separate, New table.



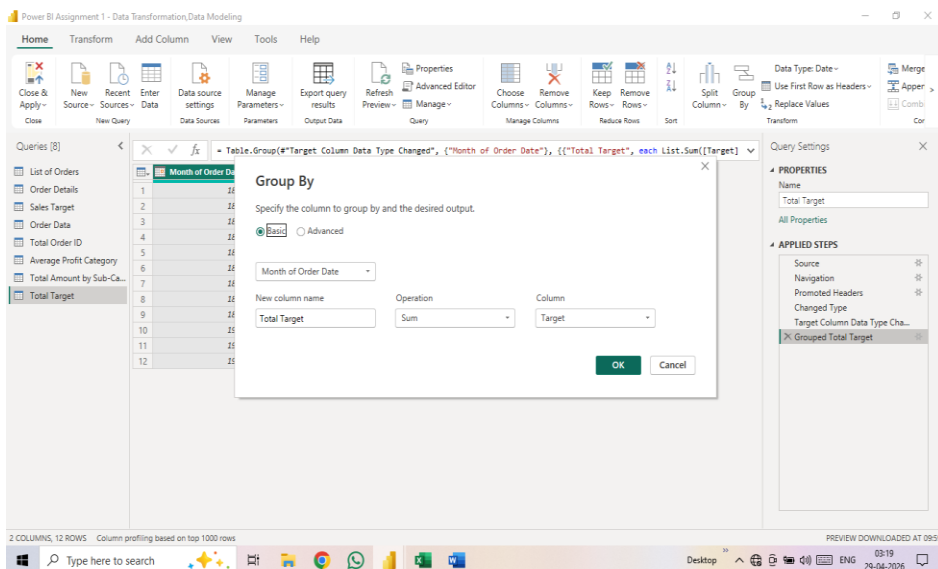
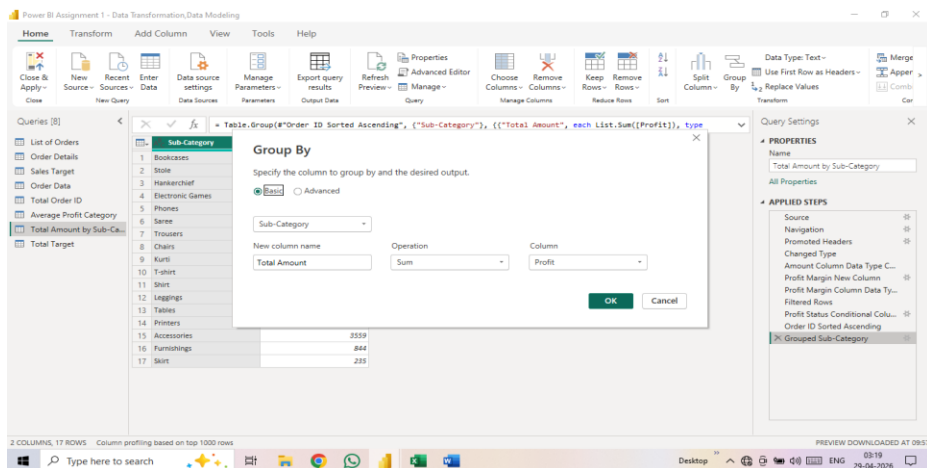
- ❖ To handle Missing values, I Utilized the “Replace Values” and “Remove Duplicates” Options.
- ❖ By Applying Sorting and Filtering, I was able to Segregate and View the data on a State- By -State Basis.



- ❖ Furthermore, I was able to sort the Data to view the order dates in Descending Order.



❖ Using the “Group By” option, I was able to Determine the total Monthly targets, the Average Total by category, and the total profits By Sub-Category.



Power BI Assignment 1 - Data Transformation, Data Modeling

Home Transform Add Column View Tools Help

Close & Apply New Source Recent Sources Enter Data Data source settings Manage Parameters Export query results Refresh Advanced Editor Choose Columns Remove Columns Keep Rows Remove Rows Split Column Group By Data Type: Text Use First Row as Headers Replace Values Merge Apper

Queries [8]

- List of Orders
- Order Details
- Sales Target
- Order Data
- Total Order ID
- Average Profit Category
- Total Amount by Sub-Ca...
- Total Target

Table.Group(#"Order ID Sorted Ascending", ("Order ID"), {"Total Order ID", each Table.RowCount(_, Int64.Type)})

Group By

Specify the column to group by and the desired output.

Basic Advanced

Order ID

New column name Operation Column

Total Order ID Count Rows

OK Cancel

Query Settings

PROPERTIES

Name

Total Order ID

APPLIED STEPS

- Source
- Navigation
- Promoted Headers
- Changed Data Type
- Amount Column Data Type C...
- Profit Margin New Column
- Profit Margin Column Data Ty...
- Profit Status Conditional Colu...
- Order ID Sorted Ascending
- X Grouped Order ID
- Count of Order ID Column Re...

2 COLUMNS, 500 ROWS Column profiling based on top 1000 rows

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Queries [8]

- List of Orders
- Order Details
- Sales Target
- Order Data
- Total Order ID
- Average Profit Category
- Total Amount by Sub-Ca...
- Total Target

Table.Group(#"Order ID Sorted Ascending", ("Category"), {"Average Profit", each List.Average({Profit}), type

Group By

Specify the column to group by and the desired output.

Basic Advanced

Category

New column name Operation Column

Average Profit Average Profit

OK Cancel

Query Settings

PROPERTIES

Name

Average Profit Category

APPLIED STEPS

- Source
- Navigation
- Promoted Headers
- Changed Data Type
- Amount Column Data Type C...
- Profit Margin New Column
- Profit Margin Column Data Ty...
- Profit Status Conditional Colu...
- Order ID Sorted Ascending
- X Grouped Average Profit

2 COLUMNS, 3 ROWS Column profiling based on top 1000 rows

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Home Transform Add Column View Tools Help

Close & Apply New Source Recent Sources Enter Data Data source settings Manage Parameters Export query results Refresh Advanced Editor Choose Columns Remove Columns Keep Rows Remove Rows Split Column Group By Data Type: Fixed decimal number Use First Row as Headers Replace Values Merge Apper

Queries [8]

- List of Orders
- Order Details
- Sales Target
- Order Data
- Total Order ID
- Average Profit Category
- Total Amount by Sub-Ca...
- Total Target

Table.TransformColumnTypes(#"Changed Data Type", {"Amount", Currency.Type})

Order ID Amount Profit Quantity Category Sub-Category

1	B-25601	1,275.00	-1148	7	Furniture	Bookcases
2	B-25601	66.00	-12	5	Clothing	Stole
3	B-25601	8.00	-2	3	Clothing	Hankchief
4	B-25601	80.00	-56	4	Electronics	Electronic Go
5	B-25602	168.00	-111	2	Electronics	Phones
6	B-25602	424.00	-272	5	Electronics	Phones
7	B-25602	2,617.00	1151	4	Electronics	Phones
8	B-25602	561.00	212	3	Clothing	Saree
9	B-25602	119.00	-5	8	Clothing	Saree
10	B-25603	1,355.00	-60	5	Clothing	Trousers
11	B-25603	24.00	-30	1	Furniture	Chairs
12	B-25603	193.00	-166	3	Clothing	Saree
13	B-25603	180.00	5	3	Clothing	Trousers
14	B-25603	116.00	16	4	Clothing	Stole
15	B-25603	107.00	36	6	Clothing	Stole
16	B-25603	12.00	1	2	Clothing	Hankchief
17	B-25603	38.00	18	2	Clothing	Kurti
18	B-25604	65.00	17	2	Clothing	T-shirt
19	B-25604	157.00	5	9	Clothing	Saree
20	B-25605	75.00	0	7	Clothing	Saree
21	B-25606	87.00	4	2	Clothing	Shirt
22	B-25607	67.00	16	4	Clothing	Stole

Query Settings

PROPERTIES

Name

Order Details

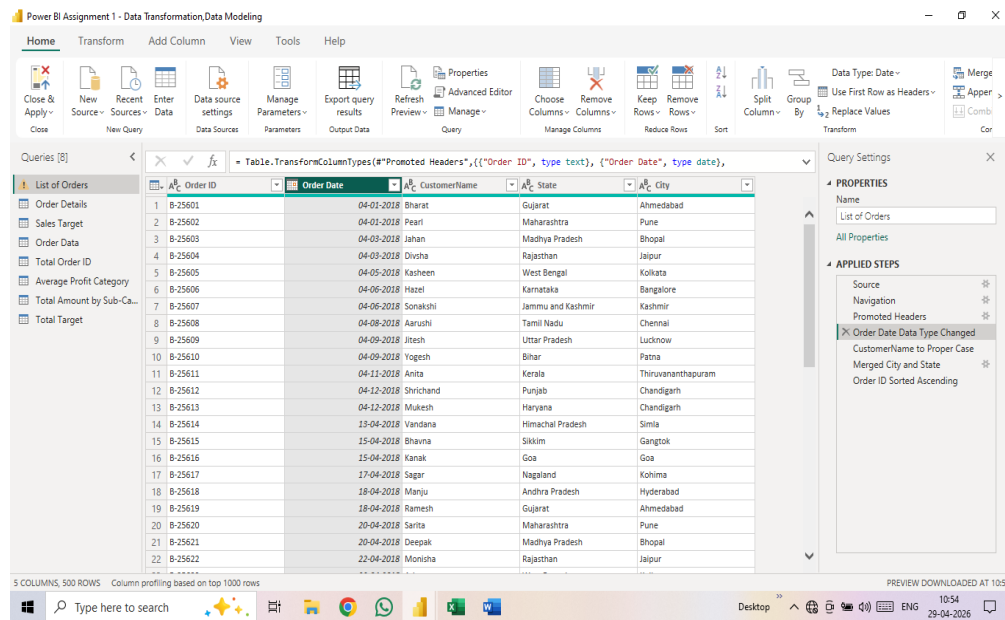
APPLIED STEPS

- Source
- Navigation
- Promoted Headers
- Changed Data Type
- X Amount Column Data Type C...
- Profit Margin New Column
- Profit Margin Column Data Ty...
- Profit Status Conditional Colu...
- Order ID Sorted Ascending

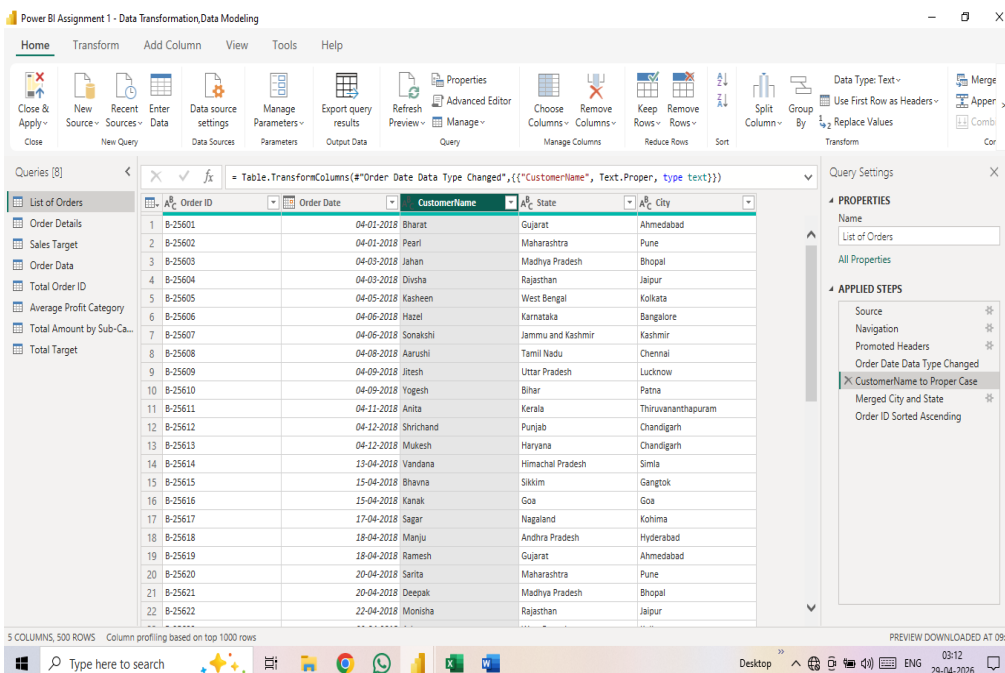
6 COLUMNS, 999+ ROWS Column profiling based on top 1000 rows

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Proper Case



Modeling Steps:

- ❖ By utilizing Data modelling, I Was Able to Establish Relationships between Separate tables.
- ❖ Furthermore , by Using the “Order ID” Found in both the “List of Orders” and “Order Details” Table.
- ❖ I was Able to Create a ONE-TO-MANY Relationship.

- ❖ Additionally, I was Able to link the “Sales Target” and “Order Details” tables based on the “Category” Field, Thereby Establishing a MANY-TO-MANY Relationship.
- ❖ Finally, By Clicking on “Manage Relationship” I Was Able to verify that These Relationships Were Indeed Active.

